MILITARY SPECIFICATION

BLANKERS, INTERFERENCE, MX-8757/UPX, MX-8758/UPX, AND MX-8758A/UPX

This amendment forms a part of Military Specification MIL-B-28850(EC), dated 13 April 1979, and is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

Page 1

TITLE: Delete "and MX-8758/UPX" and substitute ", MX-8758/UPX, and MX-8758A/UPX".

1.1, line 1: Delete "and MX-8758/UPX" and substitute ", MX-8758/UPX, and MX-8758A/UPX".

2.1, after "Issues of documents.", add: "Unless otherwise specified,".

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2.1, "STANDARDS", "MILITARY", "MIL-STD-461": Delete title and substitute:

"Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference".

2.1, "STANDARDS", "MILITARY", "MIL-STD-463": Delete title and substitute:

"Definitions And System Of Units, Electromagnetic Interference, And Electromagnetic Compatibility Technology".

2.1, "STANDARDS", "MILITARY", after "MIL-STD-1130", add:

"DoD-STD-1399, Section 300 Interface Standard For Shipboard Systems, Electric Power, Alternating Current (Metric)".

3.1, lines 1 and 2: Delete "and the MX-8758/UPX" and substitute ", the MX-8758/UPX, and the MX-8758A/UPX".

3.2, line 1: Delete "two" and substitute "three".

3.2.1.1.1b, line 2: Delete "±5 percent" and substitute "±8.5 percent".

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3.2.3, lines 2 and 4: Delete "and the MX-8758/UPX" and substitute ", the MX-8758/UPX, and the MX-8758A/UPX".

3.2.3 through 3.2.3.2.1: Renumber to 3.2.4 through 3.2.4.2.1.

* 3.2.3 and 3.2.3.2.1, added by Amendment 1 and changed by Amendment 6: Delete and substitute:

"3.2.3 MX-8758A/UPX. The MX-8758A/UPX defruiter design shall employ pulse to pulse defruiting which shall be link setting selectable to be either single, dual or, 2/3 (OR) defruiting. Defruiting shall be on a single channel per mode basis for single defruiting, and on a two channel per mode basis for dual or 2/3 (OR) defruiting.

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"3.2.3.1 Modular parameters. The modular parameters shall be as specified in a and b:

a. Modes: As specified in 3.2.1.1.1, except that Mode 4 video shall be defruited only if selected by an internal link or switch. If Mode 4 defruiting is not selected, Mode 4 video will be passed from the unit undefruited.

b. Range: The range characteristics shall be link selectable or switch selectable to be either of the following:

1. Range selection 1: When this has been selected, the defruiter MX-8758A/UPX shall defruit from mode tag to mode tag, regardless of the interlace sequence, for any repetition rate from 5 pulse repetition frequency (PRF) to 1500 PRF. The unit shall operate with any stable or staggered PRF in this range. The unit shall output undefruited video past 350 nm ±8.5 percent.

2. Range selection 2: When this has been selected, the unit shall defruit all video which has a range less than the selected maximum range. The selected maximum range shall be link selectable or internal switch selectable to be any range from 100 nm to 325 nm ±5 percent in increments of 25 nm or less. Video beyond the selected maximum range shall be undefruited output from the MX-8758A/UPX."

3.2.3 (new 3.2.4), line 4: After "(350 nm)" insert "±8.5 percent".

3.2.4: Delete.

3.2.5: Delete text and substitute: "The components of the defruiter shall conform to the requirements of MIL-STD-461, Class A4 and the interference limits specified therein. In service, the defruiter will be operated near, or interconnected with, high power radar equipment from which high levels of RF energy will be radiated and conducted to the defruiter. Provisions shall be made by incorporation of shielding or filtering, or both, in the equipment for proper operation.".

3.2.6a: Delete "2" and substitute "4".

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3.2.9: Delete "and MX-8758/UPX" and substitute ", MX-8758/UPX, and MX-8758A/UPX".

3.2.11, line 1: Delete "and the MX-8758/UPX" and substitute ", the MX-8758/UPX, and the MX-8758A/UPX".

3.2.11, line 9, after "The dimensions of the MX-8758/UPX", add: "and MX-8758A/UPX".

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3.2.12.1, line 2, after "60 Hz ±10 percent", add: "(having the characteristics of Type I power as specified in DoD-STD-1399, Section 300).".

3.2.12.1, last sentence: Delete and substitute: "The electrical requirements of DoD-STD-1399, Section 300 and MIL-E-16400F apply."

* 3.2.12.3, line 5, after "MX-8758/UPX", add "and the MX-8758A/UPX".

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3.2.21, add as last sentence: "The MX-8758A/UPX defruiter shall have a platform mount only.".

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3.2.24.1, line 1: Delete "Whenever practicable," and begin the sentence with "The".

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Add the following new paragraph 3.3.2.3 and renumber existing 3.3.2.3 through 3.3.2.17 to 3.3.2.4 through 3.3.2.18:

"3.3.2.3 MX-8758A/UPX. The MX-8758A/UPX defruiter shall perform pulse to pulse single, dual, or 2/3 (OR) defruiting, employing up to ten memory channels. When the defruiter set is turned on, the total memory need not be blank. The incoming video in response to the first interrogation is

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received from the interrogator set (or RT unit) and shall be processed (as required elsewhere in this specification), and shall be read into the appropriate mode's memory channel 1. The mode of the incoming video is determined by the presence of the appropriate mode tags. The incoming video shall be clocked into memory until the range specified in 3.2.3.1b has occurred, or until the maximum range of the defruiter is reached, (after which video shall be passed through the unit undefruited), or until another set of mode tags has been received. When video from the second interrogation of the same mode occurs, it shall be first processed and then read into the same memory channel in the same position and in the same manner as the previously stored reply video. Approximately simultaneously, the delayed video from memory channel 1 shall be read out and stored in memory channel 2 for that mode (to allow for dual defruiting and 2/3 (OR) defruiting, if selected). Mode 4 replies shall be internally selectable by link or switch to be one of the following: a) defruited as time decoded video (with separate input and output rear panel jacks from the composite video jacks) or b) defruited as reply video (this assumes another unit is doing the mode tag retiming), or c) passed through the unit undefruited for either a or b. In all other respects the operation of the MX-8758A/UPX defruiter shall be identical to the operation of the MX-8758/UPX defruiter.".

3.3.2.3f (new 3.3.2.4f), line 2: Delete "450" and substitute "1500".

3.3.2.3i (new 3.3.2.4i): Add at the beginning of the sentence "For the MX-8757/UPX and the MX-8758/UPX defruiters,".

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3.3.2.4.1c (new 3.3.2.5.1c), add as last sentence: "The MX-8758A/UPX defruiter shall not require reset signals to operate in range selection 1.".

3.3.2.4.1d (new 3.3.2.5.1d), line 2: Delete "3.3.2.4.1c" and substitute "3.3.2.5.1c". Add as last sentence, "The MX-8758A/UPX defruiter shall operate with range selection 2 if only tagged video is available.".

3.3.2.5 (new 3.3.2.6), line 2: Delete "3.3.2.4.1" and substitute "3.3.2.5.1".

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3.3.2.8 (new 3.3.2.9), line 2: After "core", insert "or random access", and line 4: Delete "450" and substitute "1500".

3.3.2.8 (new 3.3.2.9), line 5: Delete "3.3.2.3" and substitute "3.3.2.4".

3.3.2.9.1 (new 3.3.2.10.1), line 2: Delete "±5 percent" and substitute "±8.5 percent". Add as last sentence, "The range increments shall be as specified in 3.2.3.1 for the MX-8758A/UPX defruiter.".

* 3.3.2.11 (new 3.3.2.12), line 1: Delete "It is desirable for the defruiter to" and substitute "The defruiter shall". At the end of the paragraph, add "The defruiter shall also have the capability of selecting a fixed threshold adjust by use of a solderless link, or equivalent."

3.3.2.12 (new 3.3.2.13), line 2: Delete "3.3.2.12.1 through 3.3.2.12.2" and substitute "3.3.2.13.1 through 3.3.2.13.2".

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* 3.3.2.12.1.1 (new 3.3.2.13.1.1): Delete and substitute:

"3.3.2.13.1.1 Synchronous signals. Synchronous signals shall conform to the requirements specified in a through c:

a. For SIF modes, the output signal shall be full amplitude without countdown or deterioration of individual pulse characteristics for SIF reply pulses whose leading edges do not change positions relative to the last pulse (P3) of the mode identification tags or relative to the SIF mode gates by more than 0.2 μs (including internal jitter of the defruiter) from one interrogation to the next on the same mode.

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b. For Mode 4 raw video, the output signal shall be full amplitude without countdown or deterioration of individual pulse characteristics for Mode 4 raw video pulses whose leading edges do not change positions relative to the P4 pulse of the Mode 4 identification tags or relative to the Mode 4 gate by more than 0.2 μs (including internal jitter of the defruiter) from one Mode 4 interrogation to the next.

c. For Mode 4 time decoded video (TDV), the output signal shall be full amplitude without countdown or deterioration of individual pulse characteristics for Mode 4 TDV pulses whose leading edges do not change positions relative to the P4 pulse of the Mode 4 identification tags or relative to the Mode 4 gate by more than 0.4 μs (including internal jitter of defruiter) from one Mode 4 interrogation to the next.

For SIF modes and Mode 4 raw video, and TDV, the requirements specified in a through c shall apply in the presence or absence of fruit (nonsynchronous replies) which precede or overlap the reply codes."

* 3.3.2.12.1.2 (new 3.3.2.13.1.2): Delete and substitute:

"3.3.2.13.1.2 Nonsynchronous signals. Nonsynchronous signals shall conform to the requirements of a through c:

a. For SIF modes, there shall be no output for a reply pulse whose leading edge changes position relative to the last pulse (P3) of the mode identification tags or the leading edge of the SIF mode gates by more than 0.8 μs ±0.1 μs (including the internal jitter of defruiter) from one interrogation to the next on the same mode.

b. For Mode 4 raw video, there shall be no output for a reply pulse whose leading edge changes position relative to the P4 pulse of the Mode 4 identification tags or relative to the Mode 4 gate by more than 1.3 μs ±0.1 μs (including the internal jitter of defruiter) from one Mode 4 interrogation to the next.

c. For Mode 4 TDV, there shall be no output for a reply pulse whose leading edge changes position relative to the P4 pulse of the Mode 4 identification tags or relative to the Mode 4 gate by more than 1.8 μs ±0.1 μs (including the internal jitter of defruiter) from one Mode 4 interrogation to the next.

The requirements specified in a through c shall be verified for compliance with pulses of nominal duration (0.45 μs)."

3.3.2.12.2 (new 3.3.2.13.2): Add at the end of the paragraph "Dual defruiting and 2/3 (OR) defruiting and Mode 4 defruiting shall be exempt from 3.3.2.13.2.".

3.3.2.13.3, added by Amendment 1: Delete and substitute:

"3.3.2.13.3 Fruit rejection characteristics. The fruit rejection characteristics for the MX-8758A/UPX shall be given in the following formulas:

Output video PRF for single defruiting = (T2/T1)x(T2/T1)xT3
Output video PRF for dual defruiting = (T2/T1)x(T2/T1)x(T2/T1)xT3
Output video PRF for 2/3 (OR) defruiting = (2x(T2/T1))x(T2/T1)-((T2/T1)x(T2/T1)x(T2/T1)))xT3

where the following definitions apply:

The input video is uniformly distributed noise which has been pulsewidth standardized to the memory cell size, and has a minimum pulse spacing equal to the length of the accept gate.

DEAD TIME is the period of time for each pulse repetition interval (PRI) that the interrogator's receiver is off, thus allowing no video to be input to the defruiter during that time.

P1 = video PRF input-video PRF input x (DEAD TIME)x(PRF interrogation)
T1 = number of time slots in one PRI = (PRI of the interrogation - DEAD TIME)/memory cell size
T2 = number of filled time slots in one PRI = (P1)x(number of memory cells each video input pulse occupies or generates)/PRF of the interrogation
T3 = P1/((video PRF input)x(memory cell size)x(number of memory cells each video input occupies or generates))

The memory cell size for the MX-8758A/UPX shall be no greater than 0.51 μs.

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The MX-8758A/UPX shall contain at least 65,536 bits of memory for each channel for video storage. Other amounts of memory may be acceptable subject to the approval of the procuring activity.".

TABLE I, added by Amendment 4: Delete.

3.2.2.13a (new 3.3.2.14a) line 1: Delete "3.3.2.4.1a" and substitute "3.3.2.5.1a".

3.2.2.13b (new 3.3.2.14b) line 1: Delete "3.3.2.4.1b" and substitute "3.3.2.5.1b".

3.2.2.13c (new 3.3.2.14c) line 1: Delete "3.3.2.4.1c" and substitute "3.3.2.5.1c".

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3.3.2.13d (new 3.3.2.14d): In line 1, delete "3.3.2.4.1d" and substitute "3.3.2.5.1d". In line 5, delete "3.3.2.13c" and substitute "3.3.2.14c".

3.3.2.13g (new 3.3.2.14g), added by Amendment 1: Delete and substitute:

"g. A separate Mode 4 video input line shall be provided for the MX-8758A/UPX defruiter in addition to its specified inputs and outputs.".

After 3.3.2.13g (new 3.3.2.14g): Add the following:

"h. When defruiting Mode 4 TDV, no Mode 4 raw video shall be defruited, instead, it shall be routed only to the composite video output line in both standby and defruit positions. Mode 4 TDV separate input shall be outputted to only Mode 4 separate output lines in both standby and defruit positions.".

3.3.2.13.2.1.1 (new 3.3.2.14.2.1.1), line 1: Delete "3.3.2.4.2" and substitute "3.3.2.5.2".

3.3.2.13.2.1.2 (new 3.3.2.14.2.1.2), line 2: Delete "3.3.2.5" and substitute "3.3.2.6".

3.3.2.13.2.2.1e (new 3.3.2.14.2.2.1e): Delete the change made by Amendment 4.

3.3.2.13.2.2.1e (new 3.3.2.14.2.2.1e): Delete and substitute:

"e. Duration: Same as input video pulse (undelayed video pulse) +0.05 µs (with exception specified in 3.3.2.14.1)".

3.3.2.13.2.2.2 (new 3.3.2.14.2.2.2), line 2: Delete "3.3.2.5" and substitute "3.3.2.6".

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3.3.2.15 (new 3.3.2.16), second sentence (changed by Amendment 4): Delete and substitute: "Delay between either the OFF or STANDBY condition and the DEFRUIT condition for non-Mode 4 defruiting shall not exceed 1.5 µs for the MX-8757/UPX, MX-8758/UPX, and the MX-8758A/UPX.".

3.3.2.16 (new 3.3.2.17), lines 2 and 3: Delete "3.3.2.16 through 3.3.2.16.3.1" and substitute "3.3.2.17.1 through 3.3.2.17.3.1".

3.3.2.16.3 (new 3.3.2.17.3), line 3: Delete "3.3.2.16.2" and substitute "3.3.2.17.2".

3.3.2.16.3.1 (new 3.3.17.3.1), line 3: Delete "3.3.16.3" and substitute "3.3.2.17.3".

* 3.3.2.17.3.2, added by Amendment 1: Delete and substitute:

"3.3.2.17.3.2 Selection of type of defruiting. The type of defruiting (single, dual, or 2/3 (OR)) shall be link selectable."

3.3.2.17 (new 3.3.2.18), line 4: Delete "6.5.6" and substitute "6.4.6".

* 3.3.2.19, added by Amendment 1 and changed by Amendment 6: Delete and substitute:

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"3.3.2.19 MX-8758A/UPX unique requirements. The requirements in a through c are unique to the MX-8758A/UPX:

a. Future growth provisions: For future growth provisions, the MX-8758A/UPX shall have a spare printed circuit card slot. This slot shall be reserved for future possible technical improvements such as automatic thresholding, automatic pulsewidth control, automatic defruit selection, automatic accept gate control, automatic noise threshold control, and automatic adjustment of the input to memory 1 shot, and built-in test.

b. Mode 4 video storage: Each Mode 4 video pulse which passes the preliminary video processing circuitry shall be either stored to memory or generated on the output of the memory (before the logical ANDing) as a pulse which has a pulsewidth determined by whichever is longer - the original pulse's pulsewidth or the output of a retriggerable one shot which has a selectable link setting for either Mode 4 raw video or TDV.

c. Output pulse standardization: Output pulses which pass all of the other processing requirements and whose pulsewidth is from 0.3 µs to 0.5 µs shall be output as pulses which are 0.5 µs ±0.05 µs. Pulses which are greater than 0.5 µs wide shall not be modified. This feature shall be link selectable or switch selectable.".

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4.3, line 2: After "MX-8758/UPX", insert " and two sample units of the MX-8758A/UPX".

4.3, add as last sentence: "The electromagnetic compatibility (EMC) test shall be in accordance with 4.3.6.".

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4.3.1.2, line 7: Delete "3.3.2.4.2c" and substitute "3.3.2.5.2c".

4.3.1.2: At the end of the paragraph, add "for a single non-Mode 4 defruiting.".

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4.3.4.1, line 1: Delete "and MX-8758/UPX" and substitute ", MX-8758/UPX, and MX-8758A/UPX".

4.3.5, line 3: Delete "3.2.15.6" and substitute "3.2.15".

After 4.3.5, add new paragraph:

"4.3.6 EMC test. Conformance to the requirements of 3.2.5 shall be verified by utilizing the test procedure of MIL-STD-462.".

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4.4.3, lines 10 and 11: Delete and substitute: "production inspection test, shall be selected and subjected to the EMC test specified in 4.3.6.".

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4.5.2, line 2: Delete "3.2.3.1.2." and substitute "3.2.4.1.2.".

6.1, line 1: Delete "and MX-8758/UPX" and substitute ", MX-8758/UPX, and MX-8758A/UPX".

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6.4.8: At the end of the paragraph, add "The MX-8758A/UPX shall not require the reset signal to operate.".

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Add the following new paragraphs:

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"6.4.21 Dual defruiting. Dual defruiting is defruiting where the video replies from the next to last interrogation in a particular mode, the video replies from the last interrogation in the same mode, and the video replies to the present interrogation in the same mode are all logically AND'ed together.

"6.4.22 2/3 (OR) defruiting. 2/3 (OR) defruiting is defruiting in which video replies from the next to last interrogation in a particular mode is logically OR'ed to the video replies from the last interrogation in the same mode. The resulting video train is then logically AND'ed with the video replies from the present interrogation in the same mode."

Add the following new paragraph 6.5:

"6.5 Supersession data. This specification includes the requirements of AIMS 65.618A."

The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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